

QEM-8: Unified Equations for Hierarchical Mirror Lattices

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Abstract (EN):

We unify QEM-4–7 into a single field framework for hierarchical mirror lattices. Starting from a variational functional for Σ/Δ modes with a domain-wall order parameter, we derive coarse-grained equations across layers, an RG-like flow for effective couplings, and a universal causal-cone bound that collapses to the QEM-5 limit locally and to an array form in QEM-7. The coherence length scales as $\xi \sim |\theta - \theta_c|^{-1/2}$ at each level, while renormalized parameters transmit constraints upward, yielding chip-level timing, fan-out and stability windows. We provide falsifiable predictions: universality of π -echo and sech^2 envelopes under rescaling, and a layer-collapse law for the front speed. Protocols are given to measure the flow map and to verify cross-layer universality.

摘要 (ZH):

本文统一了 QEM-4–7 为适用于分层镜像晶格的单一场论框架。从具有畴壁序参量的 Σ/Δ 模变分泛函出发，我们导出了跨层粗粒化方程、有效耦合的 RG 似流以及普适的因果锥边界，后者在局部退化为 QEM-5 极限并在 QEM-7 中退化为阵列形式。相干长度在每一级按 $\xi \sim |\theta - \theta_c|^{-1/2}$ 标度，而重整化参数向上传递约束，从而给出芯片级定时、扇出和稳定性窗口。我们给出了可证伪的预测： π 回波和 sech^2 包络在标度变换下的普适性，以及层塌缩定律对前沿速度的约束。本文还给出了测量流图并验证跨层普适性的协议。

Series note: This work is part of the Quantum Entanglement Mechanics (QEM) series by Youfu Wang (2025).

1. Introduction and Motivation

QEM-4 introduced domain-wall stabilized textures; QEM-5 established a causal front for flip-waves; QEM-6 identified the Mirror-Flip Unit (MFU) as a minimal cell; QEM-7 extended to feedback-controlled arrays. QEM-8 provides unified equations that recover these limits in appropriate regimes and supply scale-bridging rules.

2. Variational Functional and Field Content

We define a Σ/Δ two-field content with a wall order parameter $m(x)$ capturing Δ -kink textures. The action $S = \int dt dx [L_\Sigma + L_\Delta + L_{\text{int}}]$ with $L_\Delta \approx \frac{1}{2} (\partial_t \varphi)^2 - \frac{1}{2} c_{\text{eff}}^2 (\partial_x \varphi)^2 - U(\varphi; \theta) + \dots$; U supports kink/antikink tanh walls and Pöschl-Teller bound states.

Coupling to Σ provides stiffness and sets c_{eff} ; dissipation γ enters via Rayleigh functional $R = \int dt dx (\gamma/2)(\partial_t \varphi)^2$.

3. Coarse-Graining and Parameter Flow

Coarse-graining from cell $\rightarrow$ tile $\rightarrow$ array yields flows for effective parameters $p=\{J, \kappa, K_{\text{inter}}, \gamma, \xi, \dots\}$.

Generic flow map: $p_{\ell+1} = F(p_{\ell}; a_{\ell})$ with monotonicity constraints ensuring stability inheritance.

ξ obeys $\xi_{\ell+1}/a_{\ell+1} \approx b \cdot (\xi_{\ell}/a_{\ell})$ under rescaling factor b ; causal bound transforms consistently.

4. Unified Causal-Cone Bound

Unified bound: $v_{\max}^{\{\ell\}} \leq (J^{\{\ell\}} \xi^{\{\ell\}})/\hbar$ with geometry factors carried by the Laplacian spectrum at level ℓ .

Local level reproduces QEM-5; array level reproduces QEM-7 with $\lambda_{\max}$ dependence.

5. Universal Textures and Echo

Near walls, standing patterns are governed by Pöschl–Teller potentials, leading to sech^2 envelopes; mirror-symmetric drives yield π -echo.

Universality claim: after rescaling by ξ and c_{eff} , envelope shapes and echo phases collapse across levels.

6. Protocols for Flow Measurement

- P1 Cross-level ξ -scaling: fit tanh walls at multiple levels; verify $\xi \sim |\theta - \theta_c|^{-1/2}$ invariant.
- P2 Cone-assembly: track fronts at cell/tile/array; confirm collapse to a unified bound curve after rescaling.
- P3 Response collapse: chirp spectroscopy across levels; rescale frequency by Ω_0 to align peaks.

7. Design Rules Across Levels

- Choose rescaling factor b and layout to keep $K_{\text{inter}}/J \leq 0.2$ at each level; maintain $\kappa/J \in [0.3, 0.5]$.
- Timing stacks must respect $\tau_\ell < 0.2 \xi^{\ell}/c_{\text{eff}}^{\ell}$ to avoid cross-level delay amplification.
- Fan-out growth limited by spectral radius of W^{ℓ} ; enforce small-world wiring to keep λ_{max} controlled.

8. Discussion and Outlook

QEM-8 supplies a single set of equations that specialize to QEM-4/5/6/7. It frames a path to hardware synthesis and suggests benchmark suites for cross-platform verification. Future work: numerical libraries and chip-layout co-design guided by the flows.

References

Lieb & Robinson (1972); Cheneau et al. (2012); Calabrese & Cardy (2006); Rajaraman (1982); Flügge (1971).

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